

CLINICAL PROBLEM-SOLVING

Caren G. Solomon, M.D., M.P.H., *Editor*

Craving Answers

Daniel O'Meara, M.D., Lea Sheward, M.D., Sydney Hartman-Munick, M.D.,
Jessica Addison, M.D., and Aya Abu-El-Haija, M.D.*In this Journal feature, information about a real patient is presented in stages (boldface type) to an expert clinician, who responds to the information by sharing relevant background and reasoning with the reader (regular type). The authors' commentary follows.*

From the Departments of Pediatrics (D.O.), Nephrology (L.S.), Adolescent Medicine (S.H.-M., J.A.), and Genetics (A.A.-E.-H.), Boston Children's Hospital, Boston. Dr. O'Meara can be contacted at daniel.omeara@childrens.harvard.edu or at the Department of Pediatrics, Boston Children's Hospital, 300 Longwood Ave., Boston, MA 02115.

Drs. Addison and Abu-El-Haija contributed equally to this article.

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A 17-year-old girl with a history of anxiety and anorexia nervosa presented to a local medical center with anxiety, malaise, lightheadedness, and palpitations. She had started an intensive outpatient nutritional rehabilitation program 2 weeks earlier and had presented to that same hospital 1 week before the current presentation with similar symptoms. At the time of that visit, her potassium level was 2.7 mmol per liter, but the results of other laboratory studies were normal. Oral potassium supplementation was provided, and she was discharged home with a prescription for 20 mEq of potassium chloride daily. However, at the time of her current presentation, she was found to have hypokalemia, hypophosphatemia, and hypomagnesemia. She received oral and parenteral electrolyte replacement and was transferred to the emergency department of an academic medical center for further care.

In general, low levels of serum electrolytes result from decreased dietary intake, intracellular shift, increased net loss from the body through gastrointestinal or renal mechanisms, or some combination thereof. This patient's history of anorexia nervosa, for which she was receiving intensive nutritional rehabilitation therapy, puts her at risk for refeeding syndrome, a potentially fatal process during which serum levels of phosphate, potassium, and magnesium may fall precipitously owing to increased intracellular demand for their presence in the performance of metabolic processes. Since refeeding syndrome typically resolves within 2 weeks, her presentation warrants assessment of her adherence to her electrolyte supplementation and consideration of other potential causes of her symptoms. Gastrointestinal causes may include self-induced vomiting, prolonged diarrhea (as may occur in response to surreptitious laxative use), and ingestions of nonedible substances such as clay, which can bind essential electrolytes within the gastrointestinal lumen. Possible renal causes of her condition include acute kidney injury associated with tubular injury and dysfunction, surreptitious use of diuretics, and acquired or inherited tubulopathies. Endocrinopathies such as diabetes mellitus and hyperaldosteronism can cause hypokalemia, hypophosphatemia, and hypomagnesemia. The patient should be asked about her urinary and bowel habits and her access to medications. A detailed family history should be obtained, along with a complete and up-to-date history of any medications the patient has taken. Her malaise, lightheadedness, and palpitations could be related to her electrolyte imbalances or to other causes, including anxiety, dehydration, orthostasis, anemia, thyroid disease, and primary cardiac dysfunction, all of which are associated with anorexia nervosa.

Two years before presentation, the patient had received a diagnosis of anorexia nervosa. At that time, she had reported fear of gaining weight and associated restrictive eating, laxative use, and hyperexercising but no purging behavior or diuretic abuse. Individual therapy that was focused on her eating disorder had led to improvement in her eating behavior and weight restoration, such that at that time her weight was 59 kg and her body-mass index (BMI [the weight in kilograms divided by the square of the height in meters]) was 22.8. However, during the Covid-19 pandemic, she struggled with mood regulation and re-engaged in caloric restriction and hyperexercise, although she stated that she was not purging or taking laxatives, diet pills, or diuretics. Consequently, in the preceding month she had lost 5.9 kg. She opted to participate in a treatment program for persons with eating disorders, which was conducted virtually owing to restrictions regarding personal contact during the pandemic. On starting the program, she immediately increased her caloric intake from <1000 kcal per day to >2000 kcal per day, which resulted in a rapid weight gain of 3.2 kg. She noted a long history of low mood, anhedonia, lethargy, hypersomnia, and feelings of excessive guilt and worthlessness, for which her primary care physician had recently prescribed sertraline and recommended psychotherapy. She took no other medications and had no other significant medical history. She reported no use of tobacco, illicit drugs, or alcohol and had no fevers, chest pain, vomiting, diarrhea, or abdominal pain.

The patient has a history of restricting-type anorexia nervosa that had responded to therapy but worsened again during the pandemic. In addition, she has symptoms related to mood that meet the criteria for major depressive disorder, making psychotherapy combined with a selective serotonin-reuptake inhibitor appropriate management. This history warrants careful screening for self-harm, suicidal ideation, and suicide plans or attempts. Her acute, marked weight loss in the preceding month, coupled with rapid weight gain in the program's first week, increases her risk of refeeding syndrome and associated complications. Unintentional causes of weight loss, such as cancer, indolent infection, and rheumatologic disease, are less likely given the correlation of her weight loss with food restriction and

high levels of exercise. The absence of reported vomiting, diarrhea, or laxative use decreases the likelihood that the cause of her electrolyte imbalances is gastrointestinal, although patients with eating disorders may not accurately report these conditions or behaviors.

On examination, the patient's temperature was 36.8°C, her blood pressure 116/93 mm Hg, her pulse 132 beats per minute, her respiratory rate 18 breaths per minute, and her oxygen saturation 100% while breathing ambient air. Her weight was 55.7 kg, and her BMI 21.5. She appeared alert and anxious. A cardiac examination showed tachycardia, with normal S1 and S2 sounds and no murmurs, rubs, or gallops. The results of pulmonary, abdominal, musculoskeletal, neurologic, and dermatologic examinations were normal. There was no edema of the sacrum or the legs, nor was there any hypertrophy of the parotid gland, dental erosion, or calloused knuckles to suggest self-induced vomiting.

The pH of a venous blood sample was 7.45. The hemoglobin level was 13.1 g per deciliter, the white-cell count 15,950 per cubic millimeter, and the platelet count 353,000 per cubic millimeter. The sodium level was 135 mmol per liter, potassium 2.55 mmol per liter, chloride 95 mmol per liter, bicarbonate 26 mmol per liter, anion gap 14 mmol per liter, blood urea nitrogen 6 mg per deciliter, and creatinine 0.4 mg per deciliter. The phosphorus level was 2.4 mg per deciliter (normal range, 2.7 to 4.9), calcium 9.2 mg per deciliter (normal range, 8.0 to 10.5), and magnesium 1.4 mg per deciliter (normal range, 1.5 to 2.2).

The results of liver-function tests and urinalysis were normal, as were the lactate level and the erythrocyte sedimentation rate. Toxicologic screens of the urine were negative for amphetamines, barbiturates, benzodiazepines, cannabinoids, cocaine, and opiates. A polymerase-chain-reaction assay for severe acute respiratory syndrome coronavirus 2 was negative. An electrocardiogram (ECG) showed sinus tachycardia with nonspecific changes in the ST-T wave and no U waves. The patient was admitted to the inpatient adolescent service for further nutritional rehabilitation, physiologic monitoring, and psychiatric assessment.

The patient has hypokalemia, hypophosphatemia, and hypomagnesemia despite supplementation,

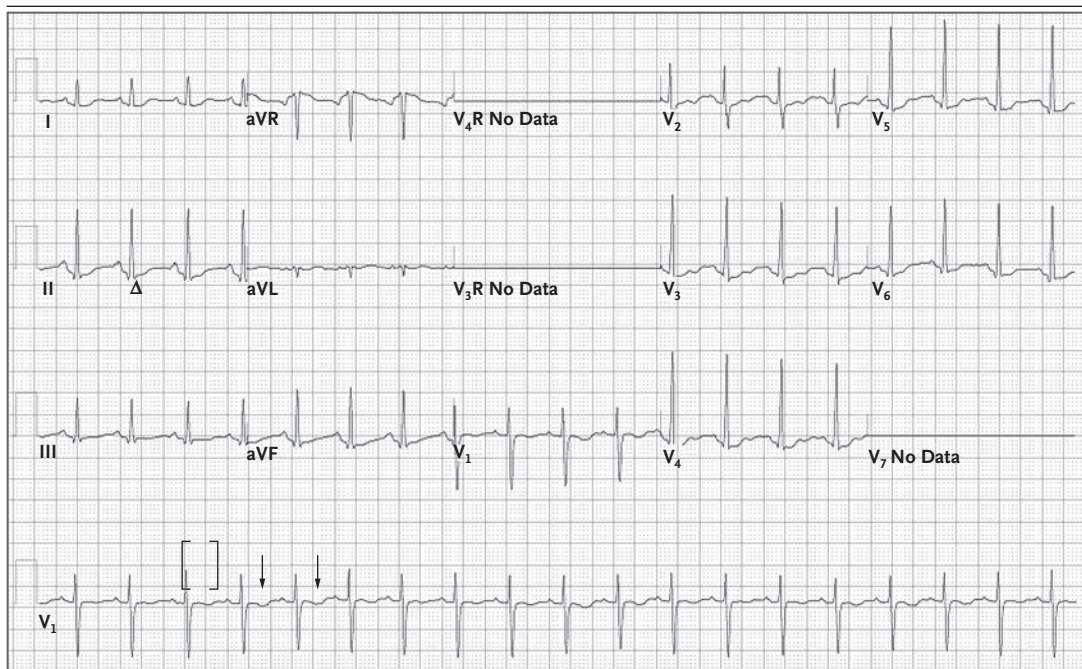


Figure 1. Representative Electrocardiograph Obtained during the Patient's Hospitalization.

Electrocardiography revealed sinus tachycardia and changes consistent with hypokalemia, including generalized ST–T wave changes with ST depression (Δ), flattening and inversion of the T wave (↓), and prolongation of the QTc interval ([]).

findings consistent with refeeding syndrome. These deficits need to be addressed, and the patient should receive maintenance electrolyte therapy to keep levels stable. The magnesium level should be watched closely, since hypomagnesemia may cause refractory hypokalemia. The elevated white-cell count with normal temperature and inflammatory markers probably represents a stress response. The findings on ECG are nonspecific; the ECG should be repeated if the electrolyte derangements persist or worsen or if further symptoms develop.

The hypophosphatemia resolved with further oral supplementation and did not recur. However, the hypokalemia persisted (lowest level, 2.51 mmol per liter), and the hypomagnesemia recurred intermittently (lowest level, 1.1 mg per deciliter) despite supervised supplementation with enteral potassium chloride of up to 60 mEq three times daily and magnesium oxide up to 800 mg three times daily. The patient also received parenteral boluses of both potassium chloride and magnesium sulfate frequently. She remained hypochloremic (lowest level, 89 mmol per liter), and hypo-

natremia developed, with sodium levels declining to 132 mmol per liter despite adequate intake and administration of isotonic intravenous fluids. Her serum bicarbonate level was intermittently elevated, reaching a high of 32 mmol per liter, with persistent mild alkalemia (the pH increased to 7.488 with venous blood gas measurement). Careful questioning and inspection of her hospital room indicated that she was not taking diuretic medication. Serial ECGs showed changes consistent with hypokalemia (Fig. 1). Given her persistent need for aggressive repletion of electrolytes, she was transferred to an intensive care step-down unit.

The resolution of hypophosphatemia with refractory hypokalemia and intermittent hypomagnesemia despite intensive repletion raises significant alarm for a process other than refeeding syndrome. This presentation is particularly concerning for renal electrolyte wasting and warrants analysis of urinary electrolytes.

Owing to concern regarding renal wasting of electrolytes, the nephrology team was consulted.

Assessment of urinary electrolytes showed a sodium level of 136 mmol per liter (expected level in hyponatremia attributable to hypovolemia, <40 mmol per liter), a potassium level of more than 100 mmol per liter (expected level in hypokalemia, <20 mmol per liter), a chloride level of 193 mmol per liter (expected level in hypochloremia, <40 mmol per liter), and a creatinine level of 31 mg per deciliter. The ratio of urinary calcium to creatinine (both measured in milligrams) was 0.06 (normal level, <0.24 mg calcium per milligram of creatinine). A subsequent urinary phosphorus level was 11.3 mg per deciliter, yielding tubular resorption of phosphate of more than 96%.

The patient shows signs of refractory hypokalemia, with extremely high urinary excretion of potassium and low urinary excretion of calcium. Measurement of her urinary electrolytes is confounded by the high doses of potassium chloride supplements she takes, but her urinary potassium, chloride, and sodium levels are nonetheless inappropriately elevated given her hypokalemia, hypochloremia, and hyponatremia attributable to hypovolemia. A laboratory diuretic screen has not been obtained but should be considered, especially given the patient's history of anorexia nervosa.

The results of renal ultrasonography were normal. The patient's serum renin activity was elevated, and the aldosterone level was normal. The results of a corticotropin stimulation test were normal. Additional history was obtained and revealed that the patient had had salt-craving behaviors since early childhood (e.g., drinking pickle juice and licking the salt off salty snacks). She also reported frequent nocturia as well as nocturnal enuresis into early adolescence. Furthermore, she reported symptomatic orthostasis, with at least one syncope episode attributed to orthostatic hypotension. Consultation with the genetics department was pursued out of concern for inherited causes of distal tubulopathies. Her family history was remarkable for a maternal aunt with unexplained hypokalemia who takes magnesium supplements. Her parents were noted to be first cousins. The family pedigree is shown in Figure S1 in the Supplementary Appendix, available with the full text of this article at NEJM.org.

The laboratory findings, together with the patient's craving for salt, the orthostasis and noc-

turia, and the family history of hypokalemia, markedly increase the suspicion for a genetic renal tubulopathy. In addition, the sharing of a common ancestor would result in large stretches of regions of homozygosity, which would in turn increase the likelihood that the patient carries a homozygous variant for a recessive condition.

The combination of hypokalemia, hypomagnesemia, and hypochloremic metabolic alkalosis is most consistent with Gitelman's syndrome, an autosomal recessive disorder affecting the thiazide-sensitive transporter encoded by *SLC12A3*. The presentation of patients with Gitelman's syndrome is often phenotypically similar to that of patients with thiazide diuretic abuse. Excessive salt loss increases delivery of sodium to the collecting duct and reabsorption by means of the aldosterone-sensitive epithelial sodium channel, which causes increased losses of urinary potassium and hydrogen as part of the body's effort to maintain electroneutrality. The initiation of a pharmacologic blockade of the distal sodium-potassium exchange with potassium-sparing diuretics, such as amiloride, spironolactone, or eplerenone, may help to ameliorate the hypokalemia if supplements alone are inadequate in maintaining acceptable serum potassium levels.

Another potential cause of the hypokalemia and hypomagnesemia is Bartter's syndrome, a genetic tubulopathy that affects the thick ascending limb of the loop of Henle and mimics the effects of loop-diuretic abuse. Although Bartter's syndrome usually causes hypercalciuria — rather than the hypocalciuria seen in this patient — in response to decreased paracellular absorption of calcium, certain subtypes of the syndrome, particularly type 3, which is caused by mutations in the chloride voltage-gated channel Kb gene (*CLCNKB*). Variants in multiple genes can lead to Bartter's syndrome, but type 3, owing to *CLCNKB*, is the most common.

Although the patient's electrolyte abnormalities are also consistent with Liddle's syndrome, an autosomal disorder affecting the aldosterone-sensitive epithelial sodium channel, the absence of hypertension, which generally appears in early childhood, is inconsistent with this diagnosis. In addition, patients with primary hyperaldosteronism would have refractory hypertension. Patients with chronic adrenal insufficiency can present with anorexia, hypotension, and salt craving, but this diagnosis is inconsistent with both the

presence of refractory hypokalemia (it typically causes hyperkalemia) and the normal results on the corticotropin stimulation test.

The patient continued to require intensive electrolyte supplementation. Given the high clinical suspicion for a hypokalemic tubulopathy and the inadequate response to supplements alone, amiloride was started at a dose of 5 mg daily, with increases in dose scheduled over 4 days with the goal of obtaining a dose of 20 mg twice daily. On this medication, the patient's potassium level normalized, and she was transitioned to oral electrolyte supplements. She was discharged in stable condition with prescriptions for magnesium oxide, potassium chloride, and sodium chloride and referred to a genetics clinic for further evaluation. While at the clinic, the family agreed to have a gene panel evaluated for the presence of a hereditary tubulopathy. The results provided the basis for a molecular diagnosis of Gitelman's syndrome owing to a homozygous pathogenic variant in *SLC12A3*, termed c.2869A→T (p.Lys947Ter). The patient continued to maintain excellent adherence to the regimen of amiloride and electrolyte supplementation, and her electrolyte levels remained stable at follow-up 15 months after her original presentation.

COMMENTARY

This 17-year-old girl with anorexia nervosa, depression, and anxiety presented with profound hypokalemia, hypophosphatemia, and hypomagnesemia after entering a nutritional rehabilitation program in response to a concern regarding rapid weight loss. The persistence of electrolyte abnormalities despite vigorous supplementation that exceeded the usual course of treatment for refeeding syndrome led clinicians to obtain a more detailed history and to pursue additional laboratory testing, the results of which were a diagnosis of Gitelman's syndrome.

Rapid, clinically significant weight loss and subsequent rapid weight regain, as occurred in this patient, are recognized risk factors for refeeding syndrome. Malnutrition resulting from rapid weight loss leads to a catabolic state that is largely mediated by a hormonal response that coincides with a decline in insulin levels and a rise in cortisol and glucagon levels. In response, the body increases glycogenolysis, lipolysis, and

protein catabolism to make ketones available to the brain for energy.¹⁻⁵ Subsequently, there is relative depletion of fat, protein, minerals, vitamins, and intracellular electrolytes. In response to the marked increase in food intake associated with nutritional rehabilitation, insulin is released, promoting glycogen and protein synthesis, with intracellular uptake of electrolytes, including phosphate, magnesium, and potassium and associated declines in the serum levels of these electrolytes, as noted in the present case. However, refeeding syndrome could not account for the patient's persistent and refractory hypokalemia and hypomagnesemia. Consequently, the differential diagnosis was broadened to include a variety of the intrinsic and extrinsic mechanisms associated with gastrointestinal and renal loss of electrolytes.

Gitelman's syndrome is caused by homozygous or compound heterozygous pathogenic variants in *SLC12A3*, the gene encoding the thiazide-sensitive sodium chloride cotransporter (Fig. 2).⁶ The syndrome is considered to be the most frequently occurring form of tubulopathy, with an estimated carrier frequency of 1:100. The relatively high prevalence of heterozygous carriers may be attributable to a selective advantage of reduced cardiovascular risk associated with low baseline blood pressure.⁷ Although this patient initially presented with elevated diastolic blood pressure, which was thought to be related to anxiety, her history of orthostatic syncopal episodes was consistent with the hypotension typical of Gitelman's syndrome. The syndrome occurs primarily in adolescents and adults but may also occur in children as early as the neonatal period. Hallmark clinical features include salt craving, fatigue, muscle weakness, dizziness, nocturia, and palpitations.⁸ (Table S2 in the Supplementary Appendix compares the clinical features of this syndrome with those of other conditions.) Typical laboratory manifestations include salt wasting, hypokalemic metabolic alkalosis, and hypocalciuria. Although hypophosphatemia and decreased reabsorption of phosphate by the proximal tubules have been observed in Gitelman's syndrome,⁹ this patient's appropriately high tubular reabsorption of phosphate and brisk response to phosphate repletion suggest that refeeding syndrome contributed to her presentation.

Hypokalemia, which can be severe, is primar-

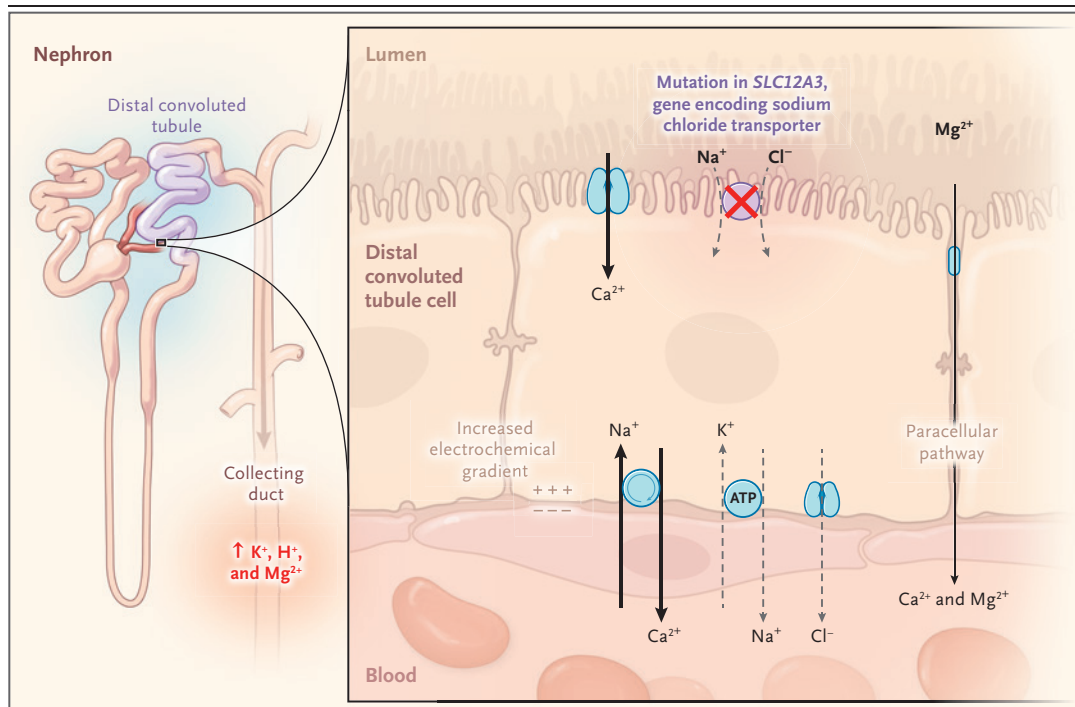


Figure 2. Deficits in Electrolyte Transport from the Renal Tubular Lumen in the Early Distal Convoluted Tubule, Accounting for Gitelman's Syndrome.

Gitelman's syndrome is caused by mutations in *SLC12A3*, the gene encoding the thiazide-sensitive sodium chloride cotransporter. Without the activity of this cotransporter, there is a lack of reabsorption of sodium and chloride at this site and increased delivery of these solutes to the collecting duct, where sodium is reabsorbed through the aldosterone-sensitive epithelial sodium channel and thereby contributes to increased urinary losses of potassium and hydrogen to maintain electroneutrality. Hypocalciuria results, in part, from an increased electrochemical gradient for the basolateral exchange of calcium (Ca^{2+}) with sodium (Na^+). Finally, hypermagnesuria results from the disrupted transcellular transport of magnesium, primarily in the collecting duct.

ily caused by hyperreninemic hyperaldosteronism, which occurs as a physiologic response to chronic salt wasting. The unexpectedly normal level of serum aldosterone in this patient may be explained by the extensive volume and electrolyte resuscitation provided at the time of assessment. As in this case, the serum potassium level typically increases and the urinary potassium level decreases after the administration of amiloride, a potassium-sparing diuretic that acts on epithelial sodium channels. Direct effects of hypomagnesemia can also contribute to hypokalemia. Normomagnesemia supports normal sodium–potassium ATPase function, which promotes intracellular potassium intake and limits the excretion of distal tubular potassium by inhibiting the outer renal medullary potassium channel.^{10,11} Although hypomagnesemia is not typically associated with Bartter's syndrome, it

does occur in some cases, particularly in late-onset cases resulting from pathogenic variants of *CLCNKB* (leading to Bartter's syndrome type 3).

Because multiple inherited tubulopathies manifest in similar ways, molecular genetic testing is necessary to confirm the diagnosis. Among patients with a clinical diagnosis of an inherited tubulopathy, next-generation sequencing and gene panels can identify a molecular diagnosis in approximately 30% of adult cases¹² and 64% of pediatric cases.¹³ In our patient, genetic testing revealed the homozygous variant c.2869A→T (p.Lys947Ter) in *SLC12A3* (ClinVar accession number, VCV000933654), which had previously been identified in a patient for whom there had been a clinical suspicion of Gitelman's syndrome.¹⁴ This is a nonsense variant that is not present in a large population cohort¹⁵ and is predicted to result in nonsense-mediated decay or protein

truncation in a gene for which loss-of-function is a known mechanism of disease. Consequently, the variant was viewed as being pathogenic.

Successful diagnosis and treatment of Gitelman's syndrome is essential to the provision of appropriate genetic counseling and the prevention of potentially life-threatening complications, such as ventricular tachycardia, seizures, and rhabdomyolysis.⁸ Education and support must be provided regarding the predictable challenges patients often face in maintaining adherence to the use of amiloride and electrolyte supplements,

including cost, transition to adulthood, and adverse effects, particularly the diarrhea caused by magnesium.⁸ The successful diagnosis and care of our patient depended on obtaining a careful history from her and her family, openness to considering alternative explanations for her electrolyte deficiencies, and close collaboration among multiple subspecialists.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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